Calculation Cover Sheet

1. Document Title: Comparing Dose Rates for Carbon vs. Stainless Steel CCC		2. System Number: PT01
2. Document Number: CCO-CAL-0006	3. Document Revision: 0	4. Page: 1 of 30

5. Summary Description

This calculation evaluates the Criticality Control Component (CCC), the confinement boundary for the Criticality Control Overpack (CCO), to determine the effect on dose rates when comparing a CCC optionally fabricated entirely of carbon steel versus the certified CCC fabricated entirely of stainless steel. This evaluation considers normal conditions of transport (NCT) regulatory dose rate limits under exclusive use requirements, i.e., dose rates at the surface of the TRUPACT-II and HalfPACT packages, and 2 meters from the surface of the packages.

7. Software Usage

Software Name	Version
1. MCNP6	6.2.0
2. Microsoft Excel	2016
3.	
4.	
5.	

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TABLE OF REVISIONS

Revision Number	Pages Affected	Revision Description
0	All	New Issue

1.0 INTRODUCTION

The TRUPACT-II and HalfPACT package licenses are planned for revision to authorize the optional use of carbon steel to fabricate the Criticality Control Component (CCC) of the Criticality Control Overpack (CCO). The purpose of this calculation is to determine the increase in dose rate for a carbon steel CCC compared to a stainless steel CCC, if any, for both gamma and neutron emitters, to meet normal conditions of transport (NCT) regulatory dose rate requirements under exclusive use shipping requirements. Based on previous analyses,[1] hypothetical accident condition (HAC) dose rate analyses have been extensively shown to be less restrictive than NCT dose rate analyses and are, therefore, not considered in this calculaton.

This calculation documents the shielding analyses for the TRUPACT-II package carrying 14 CCC assemblies, and for the HalfPACT package carrying 7 CCC assemblies, utilizing a concentrated ^{60}Co gamma and a ^{252}Cf neutron source; related CCO components are ignored for the purpose of this calculation. NCT dose rates per 10 CFR §71.47[2] are calculated at the surface of the TRUPACT-II and HalfPACT packages, and at 2 meters from the package surfaces, utilizing the models that were used for the original CCO payload-based TRUPACT-II and HalfPACT package shielding analyses.[1]

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[1] B. A. Day, *TRUPACT-II and HalfPACT Shielding Analysis*, CHG-CAL-0001, Rev. 4, Nuclear Waste Partnership LLC, Carlsbad, NM, January 2021.

[2] Title 10, Code of Federal Regulations, Part 71 (10 CFR 71), *Packaging and Transportation of Radioactive Material*, 01-01-21 Edition.

2.0 DESCRIPTION OF SHIELDING DESIGN

2.1 TRUPACT-II and HalfPACT Packagings

The TRUPACT-II and HalfPACT packagings are not designed to provide significant gamma or neutron shielding. The packagings are comprised of an outer confinement assembly (OCA) that provides both free drop and thermal protection, and an inner containment vessel (ICV) that provides the containment boundary. Two aluminum honeycomb spacer assemblies are used within the ICV, one inside each ICV torispherical head to attenuate impact loads. The only significant difference between the TRUPACT-II and HalfPACT packagings is their height (length); the HalfPACT packaging is 30 inches shorter than the TRUPACT-II packaging (44¾ vs. 74¾ payload cavity height). A complete description of the two packagings is provided in the TRUPACT-II and HalfPACT Safety Analysis Reports (SARs).[3,4]

As illustrated in Figure 2-1 and Figure 2-2, the principal shielding characteristic of these two packagings is associated with the structural materials of the containment boundary and protective overpack. The packagings utilize two relatively thin confinement/containment vessels with shell thicknesses of 3/16 and 1/4 inch. Torispherical head thicknesses are all 1/4 inch. Both the ICV and outer confinement vessel (OCV) shells and heads are constructed of Type 304 stainless steel. The ICV has a 72⅝ inch inside diameter. With a 73⅝-inch inside diameter, 94⅜-inch outside diameter, and 121½-inch overall height for the TRUPACT-II packaging (91½-inch overall height for the HalfPACT packaging), the OCA is completely encased in polyurethane foam with a density of approximately 8¼ lb/ft^3. The outer shell of the OCA is comprised of 1/4-to-3/8-inch thick, Type 304, stainless steel that serves to protect the polyurethane foam from damage encountered during normal handling and shipping operations.

Under NCT, material and distance attenuation credit is taken for the TRUPACT-II and HalfPACT packaging ICV, OCV, and OCA shells and heads and polyurethane foam.

2.2 Criticality Control Overpack

As illustrated in Figure 2-3, the CCO is an approximately 24-inch diameter, 35-inch tall steel 55-gallon drum with a steel CCC, and plywood upper and lower dunnage assemblies. The CCC is constructed of 304/304L stainless steel (or carbon steel), 6-inch Class 150 standard blind (1.0-inch flange thickness), and slip flanges and Schedule 40 pipe (0.28-inch wall thickness). The lid of the CCC is sealed with an aramid-inorganic/nbr standard ring gasket and heavy hex head steel bolts.

Under NCT, material and distance attenuation credit is taken for the CCC components, but only distance attenuation is credited for the 55-gallon drum and dunnage assemblies.

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[3] U.S. Department of Energy, *Safety Analysis Report for the TRUPACT-II Shipping Package*, Rev. 25, USNRC Certificate of Compliance 71-9218, U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

[4] U.S. Department of Energy, *HalfPACT Safety Analysis Report*, Rev. 8, USNRC Certificate of Compliance 71-9279, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

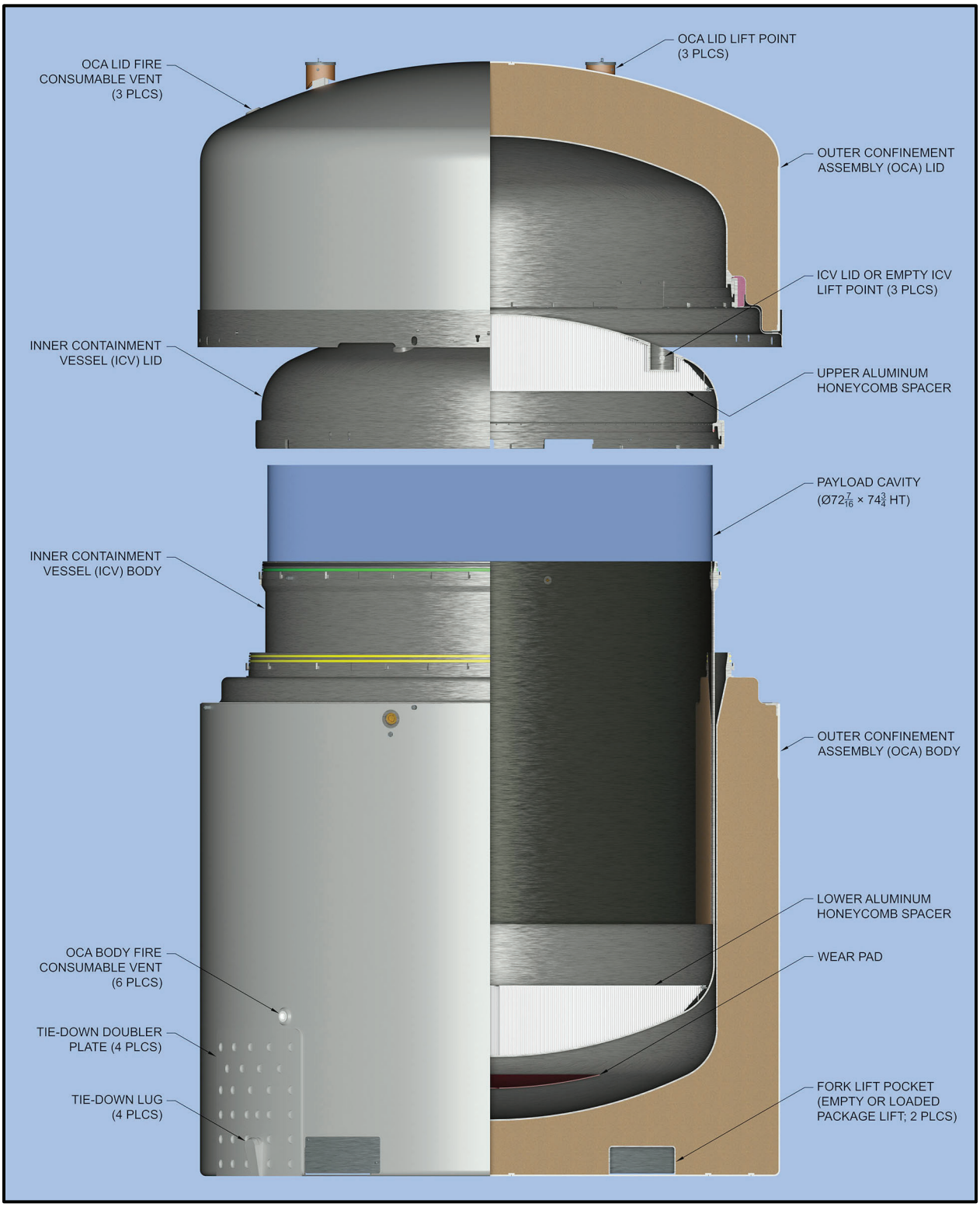

Figure 2-1 – TRUPACT-II Packaging Configuration

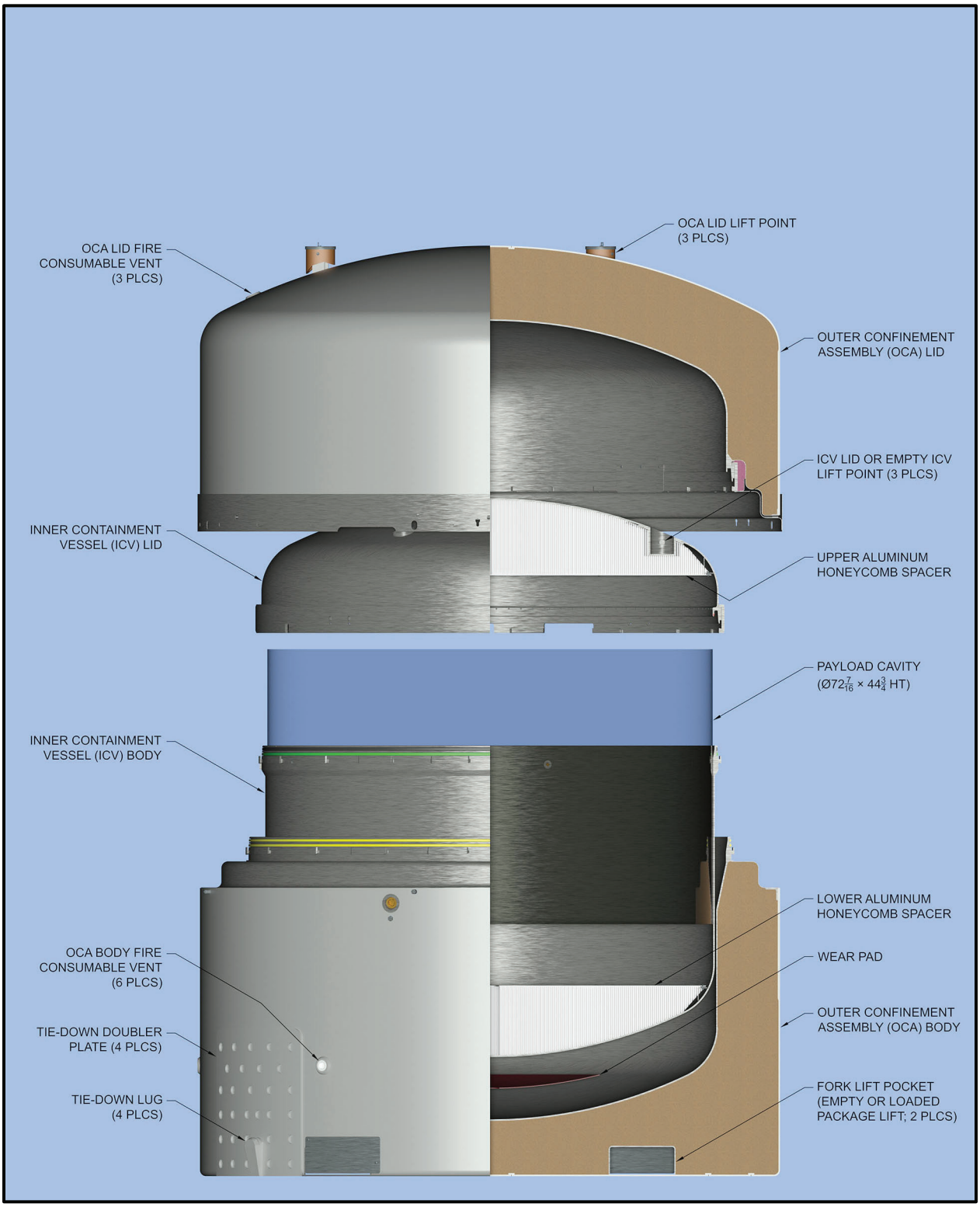

Figure 2-2 – HalfPACT Packaging Configuration

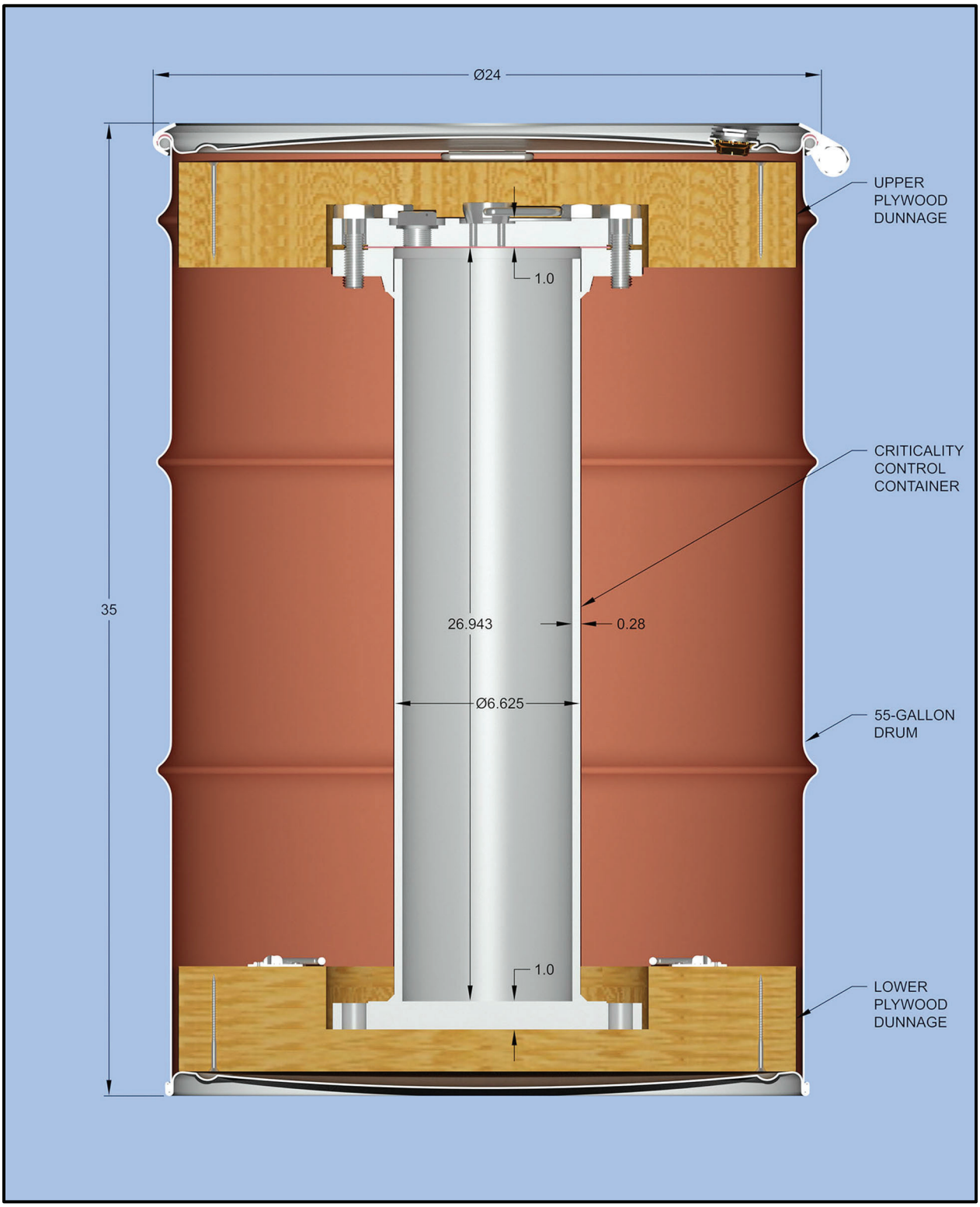

Figure 2-3 – Criticality Control Overpack

3.0 SOURCE SPECIFICATION

The gamma and neutron source terms are taken directly from Section 5.2 of the TRUPACT-II SAR.[3]

3.1 Gamma Source

The concentrated gamma source utilized for evaluation of the CCO payloads is ^{60}Co with the following photon energies and associated intensities:

Radionuclide	Photon Energy (MeV)	Intensity (%)
^{60}Co	3.469300E-01	7.600000E-03
	8.262800E-01	7.600000E-03
	1.173237E+00	9.997360E+01
	1.332501E+00	9.998560E+01
	2.158770E+00	1.110000E-03
	2.505000E+00	2.000000E-06

3.2 Neutron Source

The concentrated neutron source utilized for evaluation of the CCO payloads is ^{252}Cf (α/n plus spontaneous fission), as a function of energy per gram of material, is provided below:

Radionuclide	Neutron Energy Interval (MeV)	Neutron Source Strength (*n*/s/g)
^{252}Cf	0.0 – 0.1	0.00E+00
	0.1 – 0.5	1.80E+11
	0.5 – 1.0	2.71E+11
	1.0 – 2.0	5.36E+11
	2.0 – 3.0	4.10E+11
	3.0 – 4.0	2.73E+11
	4.0 – 6.0	2.66E+11
	6.0 – 8.0	8.53E+10
	8.0 – 10.0	2.44E+10
	10.0 – 15.0	8.36E+09

4.0 SHIELDING MODELS

The shielding models and associated materials are a copy of what is used in Section 5.3 of the TRUPACT-II SAR.[3]

4.1 Configuration of Source and Shielding

4.1.1 TRUPACT-II and HalfPACT Package Models

Damage to the packagings under NCT conditions is negligible, so the SAR drawing dimensions for the packagings are utilized. The torispherical ICV/OCV and OCA heads are treated as flat plates to maintain the TRUPACT-II overall packaging height of 120½ inches, and the HalfPACT overall packaging height of 90½ inches. The inside diameter of the 1/4-inch thick ICV shell is modeled as Ø72⅝ inches. Eliminating the gap between the ICV and OCV shells and combining the ICV shell with the 3/16-inch thick OCV shell results in an ICV/OCV outside diameter of Ø73½ inches. Rigid urethane foam with a density of 8¼ lb/ft^3 is modeled between the ICV/OCV and OCA shells. The outside diameter of the 1/4-inch thick OCA outer shell is modeled as Ø94 inches. The aluminum honeycomb end spacers are not explicitly modeled, but credit for the spacing provided by the honeycomb ensures that the radial package dose rates are the most limiting. Due to the greater material thickness of the ICV/OCV/OCA heads (as compared to the radial shells) and axial distances to detectors, only radial shielding configurations and dose rates are necessarily evaluated.

4.1.2 CCO Model

The TRUPACT-II and HalfPACT package models with a CCO payload utilize the package dimensions and materials discussed in Section 4.1.1, *TRUPACT-II and HalfPACT Package Models*. The CCO models are illustrated in Figure 4-1 through Figure 4-4. A 1 inch stub-cylinder source (i.e., 1-inch diameter × 1-inch long) is modeled at the center of each CCO payload cavity.

Damage to the CCOs under NCT conditions is negligible, so the SAR drawing dimensions for the CCO are utilized. The array spacing provided by the CCO drum is utilized to spatially locate the CCCs within the TRUPACT-II and HalfPACT package models. Each CCC is centered within the CCO drum, and the 0.28-inch thick CCC body shell and 1.00-inch thick base and lid are credited for material attenuation. The CCC body shell is modeled with a 6⅝-inch outside diameter. The CCC is modeled with a 26.943-inch cavity height. The plywood upper and lower dunnage components are credited for centering the CCC within the CCO, but no credit is taken for material attenuation of the plywood, nor for the steel 55-gallon drum. Due to the greater material thickness of the CCC lid/base (as compared to the body shell) and axial distances to detectors, only radial shielding configurations and dose rates are necessarily evaluated.

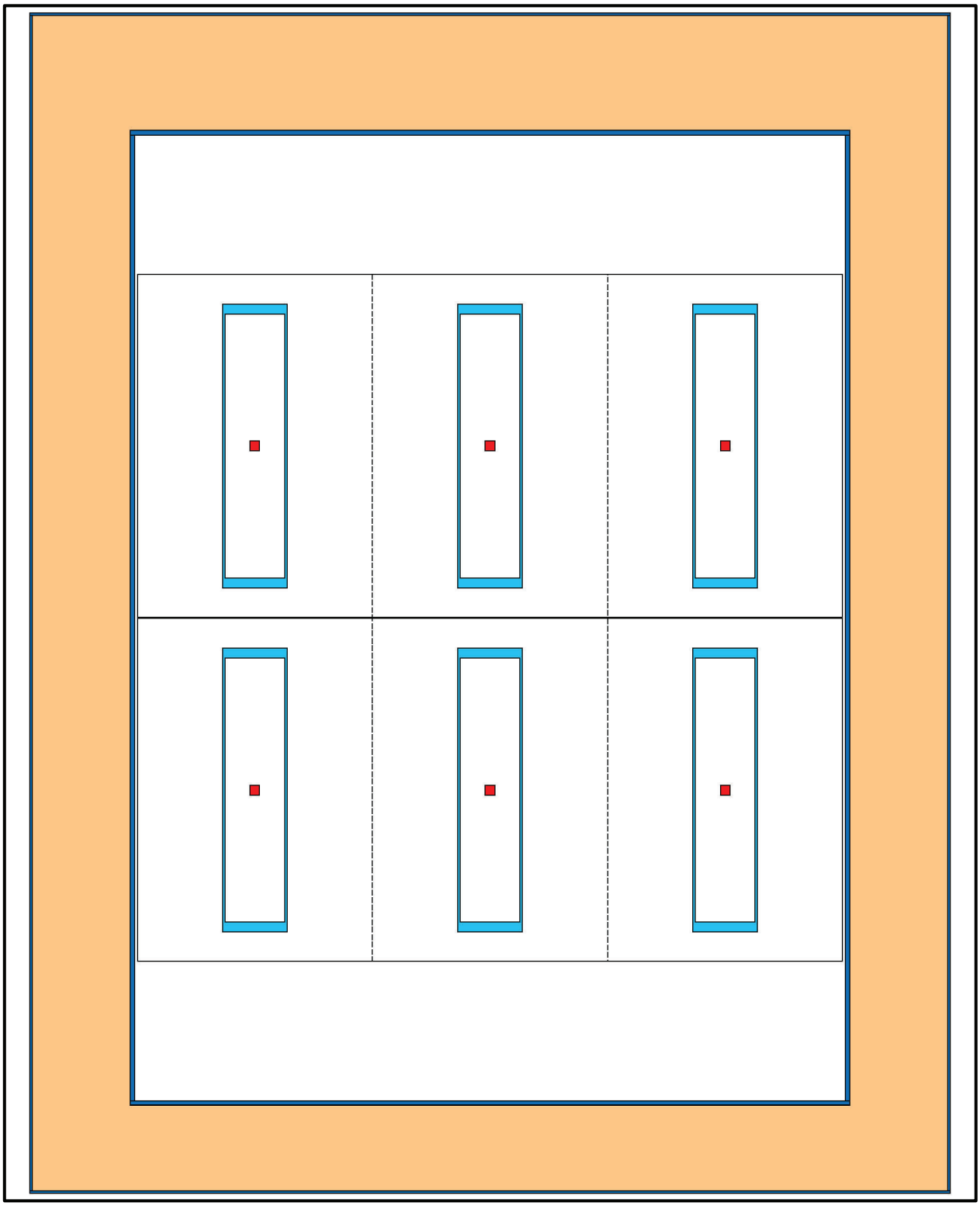

Figure 4-1 – MCNP Model of the TRUPACT-II with 14 CCOs (Side View)

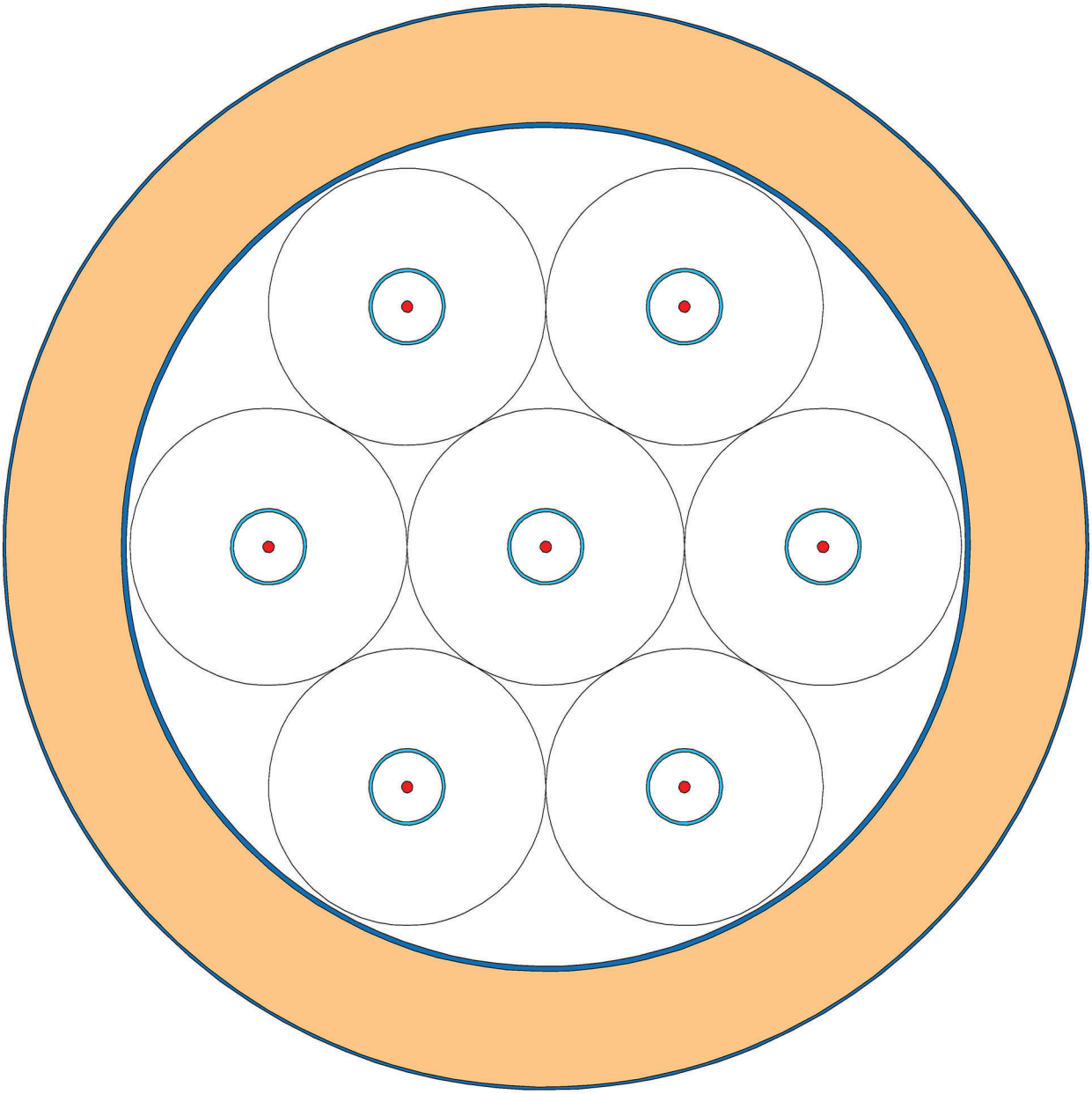

Figure 4-2 – MCNP Model of the TRUPACT-II with 14 CCOs (Top View)

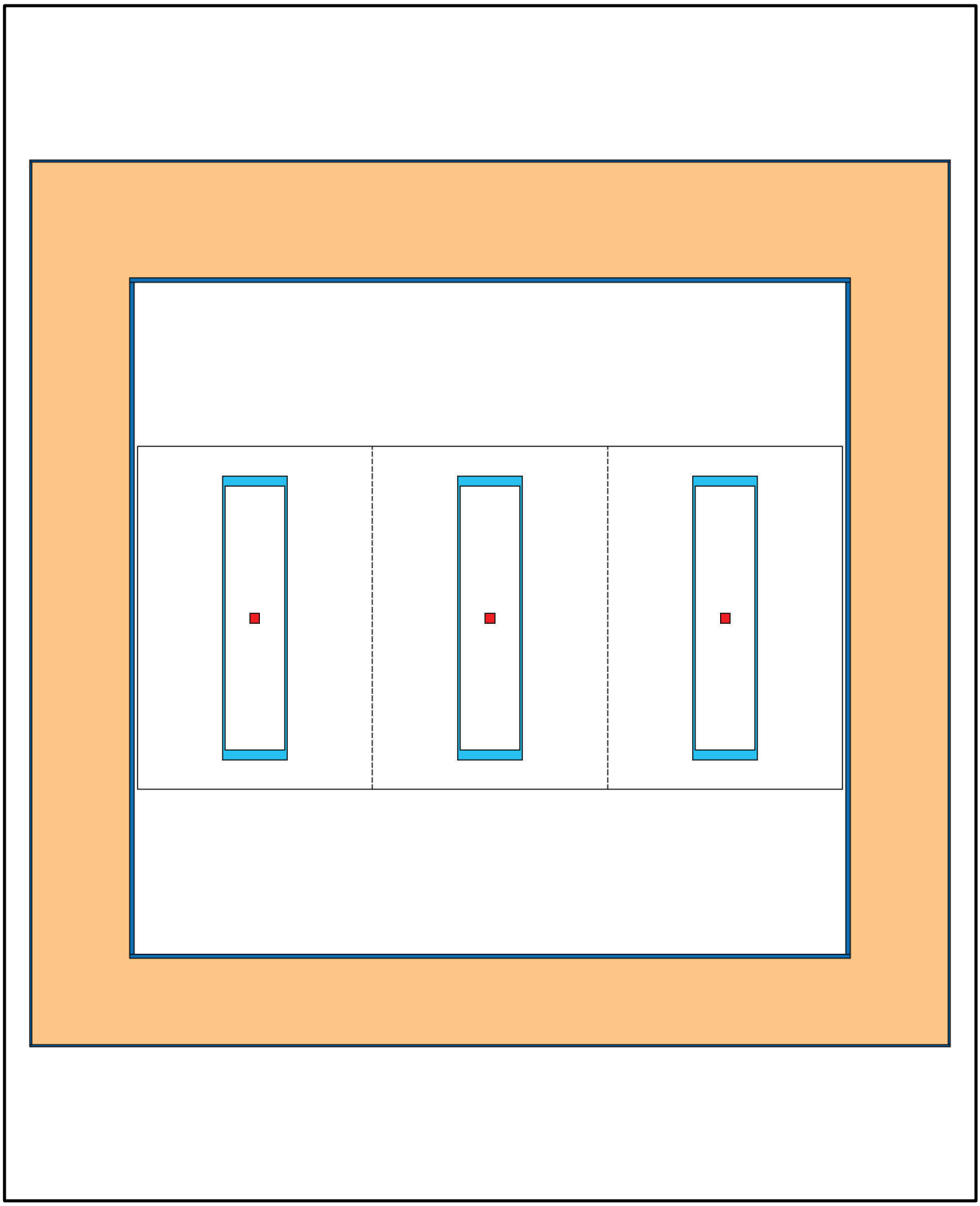

Figure 4-3 – MCNP Model of the HalfPACT with 7 CCOs (Side View)

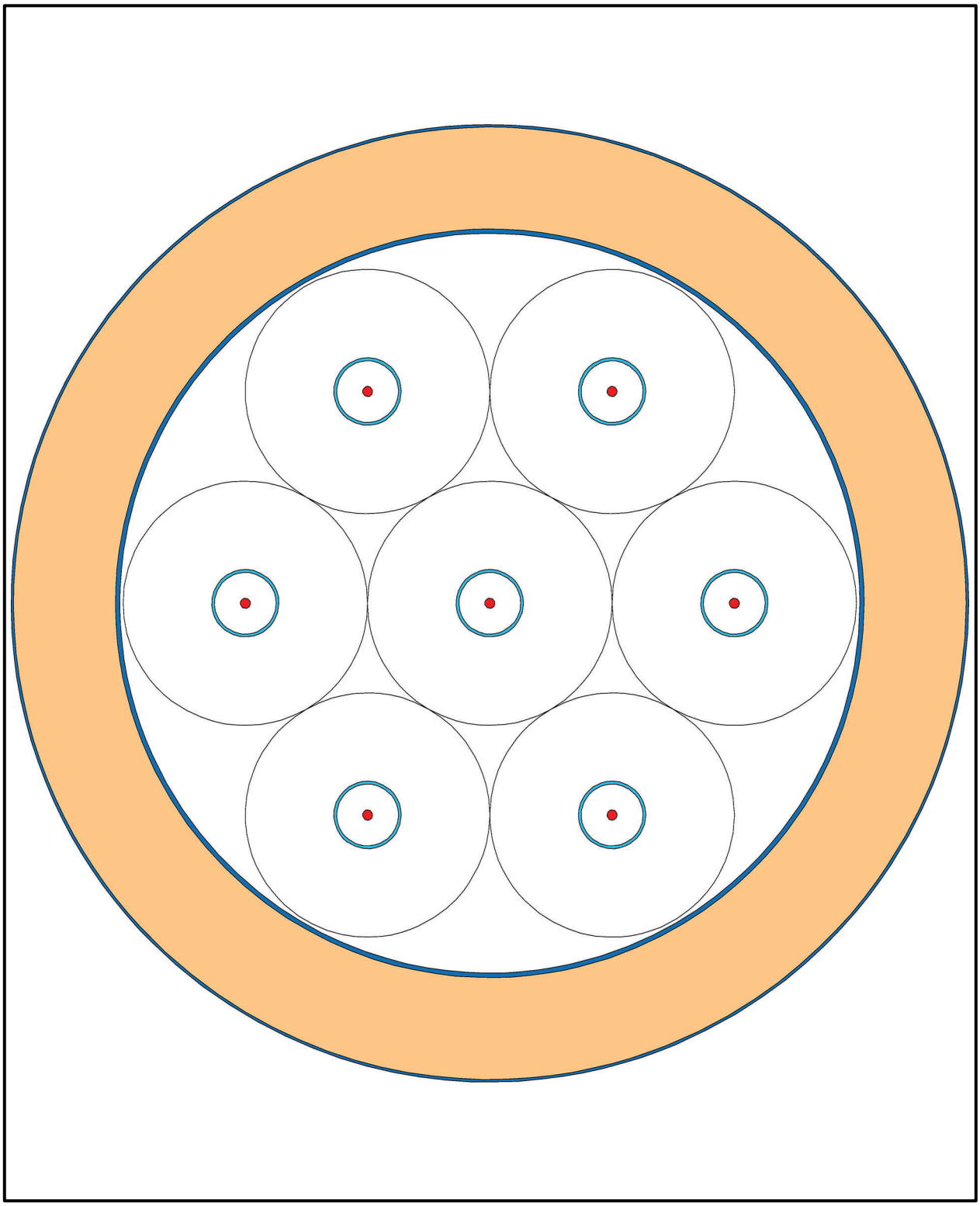

Figure 4-4 – MCNP Model of the HalfPACT with 7 CCOs (Top View)

4.2 Material Properties

The shielding models are comprised of plain carbon steel, Type 304 stainless steel, and rigid urethane foam; with material densities of 490 lb/ft³ (7.8526 g/cc), 500 lb/ft³ (8.0128 g/cc), and 8.25 lb/ft^3 (0.1322 g/cc), respectively. Table 4-1 summarizes the shield regional densities for each of these attenuating materials. Zirconium is modeled in the source region at 1.0 g/cc to represent a conservative basis for self-shielding of the concentrated 1-inch stub-cylinder source.[5]

Table 4-1 – Summary of Shield Regional Densities

	Zirconium (Source)		Carbon Steel		Type 304 Stainless Steel		Rigid Urethane Foam①	
Element	Percent	Partial Density (g/cc)	Percent	Partial Density (g/cc)	Percent	Partial Density (g/cc)	Percent	Partial Density (g/cc)
Hydrogen	—	—	—	—	—	—	7%	0.0093
Carbon	—	—	—	—	—	—	60%	0.0793
Nitrogen	—	—	—	—	—	—	8%	0.0106
Oxygen	—	—	—	—	—	—	24%	0.0317
Silicon	—	—	—	—	1%	0.0801	1%	0.0013
Chromium	—	—	—	—	19%	1.5224	—	—
Manganese	—	—	—	—	2%	0.1603	—	—
Iron	—	—	100%	7.8526	68%	5.4487	—	—
Nickel	—	—	—	—	10%	0.8013	—	—
Zirconium	100%	1.0000	—	—	—	—	—	—
Total	100%	1.0000	100%	7.8526	100%	8.0128	100%	0.1322

Notes:

① Weight percent from chemical composition (median of range) from Section 8.1.4.1.1.1, *Polyurethane Foam Chemical Composition*, of the TRUPACT-II SAR.[3]

[5] Nuclear Waste Partnership, *Self-shielding Surrogate Material Evaluation for TRUPACT-II and HalfPACT*, CHG-CAL-0002, Rev. 0, September 2012.

5.0 SHIELDING EVALUATION

5.1 Methods

MCNP6 v6.2.0 is used for the shielding analysis of the carbon steel CCCs.[6] MCNP6 is a standard, well-accepted shielding program utilized to compute dose rates for shielding licenses.

The existing three-dimensional shielding models used to certify the stainless steel CCCs per Chapter 5 of the TRUPACT-II SAR[3] capture all of the relevant design parameters of the TRUPACT-II and HalfPACT packagings and the associated CCC configurations. The only changes made to the existing shielding models was to add carbon steel as a material and change the material reference for the CCCs from stainless to carbon steel. Otherwise, the model files are identical. Dose rate results for the stainless steel CCCs are taken from Tables 5.4-7 and 5.4-8 of the TRUPACT-II SAR.

Dose rates are calculated by tallying the gamma and neutron fluxes at regions of interest and converting these fluxes to dose rates. The models are run in either photon or neutron mode, depending upon the radionuclide being evaluated.

As discussed in the original calculation,[1] combined photon/neutron mode evaluations are not employed for the neutron analyses as multiple test runs demonstrate that the gamma dose rate due to the interaction of neutrons in the shielding materials contributes a statistically insignificant (<1%) change in the dose rate as compared to the neutron dose rate. The gamma energies and intensities for ^{252}Cf are similarly ignored based on the same logic.

5.2 Input and Output Data

Two input/output cases are used to generate the results for each TRUPACT-II and HalfPACT packaging/payload combination. Case naming conventions are defined in Table 5-1.

The carbon steel CCC input files are provided in Appendix 6.1, *MCNP Input Files*. Output files are too large for inclusion in the report, so they are provided on supplemental storage media. All cases are run with a one (1) particle/second (par/s) source.

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[6] LA-UR-17-29981, *MCNP User's Manual, Code Version 6.2,* Rev. 0; Los Alamos National Laboratory, October 27, 2017.

Table 5-1 – CCO Case Naming Conventions for a Carbon Steel CCC

Case Name	Description	Case Name Variable, X	Packaging / Payload
Xng001c	Gamma, ^{60}Co, NCT	c	TRUPACT-II / CCO
Xnn001c	Neutron, ^{252}Cf, NCT	d	HalfPACT / CCO
File Extension Definitions			
Input File = .i	Output File = .o	Runtpe File = .r	Tally File = .m

5.3 Flux-to-Dose Conversion

ANSI/ANS-6.1.1-1977 flux-to-dose-rate conversion factors are utilized for both neutron and gamma radiation. These factors are provided in Table 5-2 and Table 5-3.

Table 5-2 – Neutron Flux-to-Dose Rate Conversion Factors from ANSI/ANS 6.1.1-1977

n-Energy (MeV)	DFg(E) (*n*/cm²-s to mrem/hr)	Quality Factor	*n*-Energy (MeV)	DFg(E) (*n*/cm²-s to mrem/hr)	Quality Factor
2.50E-08	3.67E-03	2.0	5.00E-01	9.26E-02	11.0
1.00E-07	3.67E-03	2.0	1.00E+00	1.32E-01	11.0
1.00E-06	4.46E-03	2.0	2.50E+00	1.25E-01	9.0
1.00E-05	4.54E-03	2.0	5.00E+00	1.56E-01	8.0
1.00E-04	4.18E-03	2.0	7.00E+00	1.47E-01	7.0
1.00E-03	3.76E-03	2.0	1.00E+01	1.47E-01	6.5
1.00E-02	3.56E-03	2.5	1.40E+01	2.08E-01	7.5
1.00E-01	2.17E-02	7.5	2.00E+01	2.27E-01	8.0

Table 5-3 – Gamma Flux-to-Dose Rate Conversion Factors from ANSI/ANS 6.1.1-1977

γ-Energy (MeV)	DFg(E) (γ/cm²-s to mrem/hr)	γ-Energy (MeV)	DFg(E) (γ/cm²-s to mrem/hr)
0.01	3.96E-03	1.40	2.51E-03
0.03	5.82E-04	1.80	2.99E-03
0.05	2.90E-04	2.20	3.42E-03
0.07	2.58E-04	2.60	3.82E-03
0.10	2.83E-04	2.80	4.01E-03
0.15	3.79E-04	3.25	4.41E-03
0.20	5.01E-04	3.75	4.83E-03
0.25	6.31E-04	4.25	5.23E-03
0.30	7.59E-04	4.75	5.60E-03
0.35	8.78E-04	5.00	5.80E-03
0.40	9.85E-04	5.25	6.01E-03
0.45	1.08E-03	5.75	6.37E-03
0.50	1.17E-03	6.25	6.74E-03
0.55	1.27E-03	6.75	7.11E-03
0.60	1.36E-03	7.50	7.66E-03
0.65	1.44E-03	9.00	8.77E-03
0.70	1.52E-03	11.00	1.03E-02
0.80	1.68E-03	13.00	1.18E-02
1.00	1.98E-03	15.00	1.33E-02

5.4 External Radiation Levels

The MCNP6 models for determining external radiation levels use segmented surface detectors to calculate the radial dose rates at the surface of the TRUPACT-II and HalfPACT package, and at 2 meters from the surface of the TRUPACT-II and HalfPACT packages for NCT. The detectors are axially aligned with the centerline halfway between the sources in the lower and upper CCCs for the TRUPACT-II package, and axially aligned with the sources for the HalfPACT package to report the maximum dose rate.

Dose rates and the associated statistical uncertainty (i.e., tally error) are computed based on a 1 par/s source strength. Conservatively accounting for the tally error, dose rates at the surface of

the packages and two meters from the surface of the package are compared for the optional carbon steel CCCs versus the certified stainless steel CCCs.

The dose rates and tally errors computed for 14 CCCs within a TRUPACT-II package, and 7 CCCs within a HalfPACT package with concentrated gamma and neutron sources having a 1 par/s source strength, are provided in Table 5-4 and Table 5-5, respectively. The stainless steel CCC results come from Table 5.4-7 and Table 5.4-8 of the TRUPACT-II SAR.[3]

Table 5-4 – TRUPACT-II – Carbon vs. Stainless Steel CCC Comparison

NCT Tally Location	Stainless Steel (SS) CCCs			Carbon Steel (CS) CCCs			Dose Rate Change — CS vs. SS
	Calculated Dose Rate (mrem/hr / par/s)	Tally Error	Adjusted Dose Rate (mrem/hr / par/s)	Calculated Dose Rate (mrem/hr / par/s)	Tally Error	Adjusted Dose Rate (mrem/hr / par/s)	
Results for ^{60}Co Gamma Sources							
Surface	5.78E-09	0.2%	5.79E-09	5.82E-09	0.2%	5.83E-09	+0.69%
2-Meters	8.31E-10	0.2%	8.33E-10	8.36E-10	0.2%	8.38E-10	+0.60%
Results for ^{252}Cf Neutron Sources							
Surface	5.69E-07	0.2%	5.70E-07	5.70E-07	0.2%	5.71E-07	+0.18%
2-Meters	7.56E-08	0.2%	7.58E-08	7.59E-08	0.2%	7.61E-08	+0.40%

Table 5-5 – HalfPACT – Carbon vs. Stainless Steel CCC Comparison

NCT Tally Location	Stainless Steel (SS) CCCs			Carbon Steel (CS) CCCs			Dose Rate Change — CS vs. SS
	Calculated Dose Rate (mrem/hr / par/s)	Tally Error	Adjusted Dose Rate (mrem/hr / par/s)	Calculated Dose Rate (mrem/hr / par/s)	Tally Error	Adjusted Dose Rate (mrem/hr / par/s)	
Results for ^{60}Co Gamma Sources							
Surface	8.48E-09	0.1%	8.49E-09	8.52E-09	0.1%	8.53E-09	+0.47%
2-Meters	8.51E-10	0.2%	8.53E-10	8.56E-10	0.2%	8.58E-10	+0.59%
Results for ^{252}Cf Neutron Sources							
Surface	7.96E-07	0.2%	7.98E-07	7.96E-07	0.1%	7.97E-07	–0.13%
2-Meters	7.57E-08	0.2%	7.59E-08	7.60E-08	0.2%	7.61E-08	+0.26%

5.5 Summary

Based on the geometry, materials of construction, configuration of the TRUPACT-II and HalfPACT packagings, and configuration of the CCO assembly, the maximum 0.69% increase in dose rate for the optional carbon steel CCCs versus the certified stainless steel CCCs will necessitate increasing the 10% administrative margin discussed in Paragraph 10 of Section 5.5.10 of the TRUPACT-II SAR[3]. The increase is from 10% to 11 % to account for the aforementioned small dose rate increase for CCCs fabricated with the optional carbon steel, as well as any uncertainties associated with this evaluation. Thus, the combined sum of the sum of fractions for the gamma and the neutron source term shall be less than or equal to 0.89.

6.0 APPENDICES

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6.1.2.2 Carbon Steel CCCs with ^{252}Cf Sources – dnn001c.i 29

6.1 MCNP Input Files

6.1.1 TRUPACT-II Package Models

6.1.1.1 Carbon Steel CCCs with ^{60}Co Sources – cng001c.i

```
title TRUPACT-II CCO NCT Gamma Concentrated 1 g/cc Co60; CS CCC
C
C Cell Cards
C
1   1 -1        10  -11  -14                u=1 imp:p=1  $ Lower Source
2   4 -7.85260  30  -32  -34 (-31:33:35)   u=1 imp:p=1  $ Lower CCC
3   0           31  -33  -35 (-10:11:14)   u=1 imp:p=1  $ Lower CCC cavity
4   0           #1 #2 #3                    u=1 imp:p=1  $ Lower Drum cavity
5   1 -1        20  -21  -24                u=2 imp:p=1  $ Upper Source
6   4 -7.85260  30  -32  -34 (-31:33:35)   u=2 imp:p=1  $ Upper CCC
7   0           31  -33  -35 (-20:21:24)   u=2 imp:p=1  $ Upper CCC cavity
8   0           #5 #6 #7                    u=2 imp:p=1  $ Upper Drum cavity
11  0           50  -51  -52     trcl=1 fill=1 imp:p=1  $ CCO #1
12  0           50  -51  -52     trcl=2 fill=1 imp:p=1  $ CCO #2
13  0           50  -51  -52     trcl=3 fill=1 imp:p=1  $ CCO #3
14  0           50  -51  -52     trcl=4 fill=1 imp:p=1  $ CCO #4
15  0           50  -51  -52     trcl=5 fill=1 imp:p=1  $ CCO #5
16  0           50  -51  -52     trcl=6 fill=1 imp:p=1  $ CCO #6
17  0           50  -51  -52     trcl=7 fill=1 imp:p=1  $ CCO #7
18  0           50  -51  -52     trcl=8 fill=2 imp:p=1  $ CCO #8
19  0           50  -51  -52     trcl=9 fill=2 imp:p=1  $ CCO #9
20  0           50  -51  -52    trcl=10 fill=2 imp:p=1  $ CCO #10
21  0           50  -51  -52    trcl=11 fill=2 imp:p=1  $ CCO #11
22  0           50  -51  -52    trcl=12 fill=2 imp:p=1  $ CCO #12
23  0           50  -51  -52    trcl=13 fill=2 imp:p=1  $ CCO #13
24  0           50  -51  -52    trcl=14 fill=2 imp:p=1  $ CCO #14
200 0           210 -250 -230
                #11 #12 #13 #14 #15 #16 #17
                #18 #19 #20 #21 #22 #23 #24    imp:p=1  $ ICV cavity
210 2 -8.01280  200 -210 -220                  imp:p=1  $ ICV/OCV bottom
220 2 -8.01280  250 -240 -220                  imp:p=1  $ ICV/OCV top
230 2 -8.01280  210 -250  230 -220             imp:p=4  $ ICV/OCV shell
240 3 -0.13215  110 -150 -130 (-200:220:240)   imp:p=8  $ OCA annulus
250 2 -8.01280  100 -110 -120                  imp:p=1  $ OCA bottom
260 2 -8.01280  150 -140 -120                  imp:p=1  $ OCA top
270 2 -8.01280  110 -150  130 -120             imp:p=16 $ OCA shell
777 0           120 -160  100 -140             imp:p=32 $ Outside to tally
888 0          -999 (-100:160:140)             imp:p=8  $ Outside pkg
999 0           999                            imp:p=0  $ Outside world

C
C Geometry Cards
C
10  pz 147.955                                          $ Lower Source bottom
11  pz 150.495                                          $ Lower Source top
14  c/z 0 0 1.27                                        $ Lower Source radius
20  pz 147.955                                          $ Upper Source bottom
21  pz 150.495                                          $ Upper Source top
24  c/z 0 0 1.27                                        $ Upper Source radius
C
30  pz 112.467                                          $ CCC base (outside)
31  pz 115.007                                          $ CCC base (inside)
32  pz 185.983                                          $ CCC lid (outside)
33  pz 183.443                                          $ CCC lid (inside)
34  cz 8.414                                            $ CCC shell (outside)
35  cz 7.703                                            $ CCC shell (inside)
C
50  pz 104.775                                          $ Drum bottom
51  pz 193.675                                          $ Drum top
52  cz 30.48                                            $ Drum radius (outside)
C
100 pz 0                                                $ OCA bottom (outside)
110 pz 0.635                                            $ OCA bottom (inside)
120 cz 119.38                                           $ OCA shell (outside)
130 cz 118.745                                          $ OCA shell (inside)
140 pz 306.07                                           $ OCA top (outside)
150 pz 305.435                                          $ OCA top (inside)
160 cz 319.38                                           $ 2-meter from OCA
C
```

```
200 pz 22.86                                          $ ICV/OCV bottom (outside)
210 pz 24.13                                          $ ICV/OCV bottom (inside)
220 cz 93.345                                         $ ICV/OCV shell (outside)
230 cz 92.234                                         $ ICV/OCV shell (inside)
240 pz 275.59                                         $ ICV/OCV top (outside)
250 pz 274.32                                         $ ICV/OCV top (inside)
C
777 pz 146.685                                        $ Ring tally (lower)
888 pz 151.765                                        $ Ring tally (upper)
999 sz 149.225 446.38                                 $ Outside world

C
C Translation Cards
C
*tr1 0 0 -44.609
*tr2 52.793 30.48 -44.609
*tr3 0 60.96 -44.609
*tr4 -52.793 30.48 -44.609
*tr5 -52.793 -30.48 -44.609
*tr6 0 -60.96 -44.609
*tr7 52.793 -30.48 -44.609
*tr8 0 0 44.609
*tr9 52.793 30.48 44.609
*tr10 0 60.96 44.609
*tr11 -52.793 30.48 44.609
*tr12 -52.793 -30.48 44.609
*tr13 0 -60.96 44.609
*tr14 52.793 -30.48 44.609
C
C Physics Cards
C
mode p
C
C Source Cards
C
C Cylindrical Sources
sdef pos=d1 erg=d2 par=2 axs=0 0 1 rad=d3 ext=d4
sc1 Co60
C Source Position and Fraction
si1 L 0 0 104.616
      52.793 30.48 104.616
      0 60.96 104.616
      -52.793 30.48 104.616
      -52.793 -30.48 104.616
      0 -60.96 104.616
      52.793 -30.48 104.616
      0 0 193.834
      52.793 30.48 193.834
      0 60.96 193.834
      -52.793 30.48 193.834
      -52.793 -30.48 193.834
      0 -60.96 193.834
      52.793 -30.48 193.834
sp1   1 1 1 1 1 1 1 1 1 1 1 1 1 1
C Source Energy (MeV) and Intensity (%)
si2 L 3.469300E-01 8.262800E-01 1.173237E+00 1.332501E+00
     2.158770E+00 2.505000E+00
sp2 7.600000E-03 7.600000E-03 9.997360E+01 9.998560E+01
     1.110000E-03 2.000000E-06
C Source Radius
si3 0 1.27
sp3 -21 1
C Source Extent
si4 -1.27 1.27
sp4 -21 0
C
C Tally Cards
C
f2:p 120
fc2 Surface Dose (mrem/hr)                            $ Surface detector
fs2 888 777 t
sd2 1.0 3810.44 1.0 1.0                               $ Surface tally area
f22:p 160
fc22 2-meter Dose (mrem/hr)                           $ 2-m detector
fs22 888 777 t
sd22 1.0 10194.144 1.0 1.0                            $ Surface tally area
C
C ANSI/ANS-6.1.1-1977 Gamma Flux to Dose Factors (mrem/hr)
de0 1.0e-02 3.0e-02 5.0e-02 7.0e-02 1.0e-01           $ Energy (Mev)
     1.5e-01 2.0e-01 2.5e-01 3.0e-01 3.5e-01
     4.0e-01 4.5e-01 5.0e-01 5.5e-01 6.0e-01
```

```
     6.5e-01 7.0e-01 8.0e-01 1.00 1.40
     1.80 2.20 2.60 2.80 3.25
     3.75 4.25 4.75 5.00 5.25
     5.75 6.25 6.75 7.50 9.00
     11.00 13.00 15.0000
df0 3.96e-03 5.82e-04 2.90e-04 2.58e-04 2.83e-04      $ Conversion (mrem/hr)
     3.79e-04 5.01e-04 6.31e-04 7.59e-04 8.78e-04
     9.85e-04 1.08e-03 1.17e-03 1.27e-03 1.36e-03
     1.44e-03 1.52e-03 1.68e-03 1.98e-03 2.51e-03
     2.99e-03 3.42e-03 3.82e-03 4.01e-03 4.41e-03
     4.83e-03 5.23e-03 5.60e-03 5.80e-03 6.01e-03
     6.37e-03 6.74e-03 7.11e-03 7.66e-03 8.77e-03
     1.03e-02 1.18e-02 1.33e-02
C
C Material Cards
C
m1   40000 -1.00                                      $ Zirconium
m2   14000 -0.01                                      $ Stainless Steel
     24000 -0.19
     25000 -0.02
     26000 -0.68
     28000 -0.10
m3   6000  -0.60                                      $ Urethane Foam
     7000  -0.08
     8000  -0.24
     1000  -0.07
     14000 -0.01
m4   26000 -1.00                                      $ Carbon Steel (iron)
C
C Runtime and Print Cards
C
prdmp j j 1 2
ctme 30
```

6.1.1.2 Carbon Steel CCCs with ^{252}Cf Sources – cnn001c.i

```
title TRUPACT-II CCO NCT Neutron Concentrated 1 g/cc Cf252; CS CCC
C
C Cell Cards
C
1   1 -1        10  -11  -14                u=1 imp:n=1  $ Lower Source
2   4 -7.85260  30  -32  -34 (-31:33:35)    u=1 imp:n=1  $ Lower CCC
3   0           31  -33  -35 (-10:11:14)    u=1 imp:n=1  $ Lower CCC cavity
4   0           #1 #2 #3                    u=1 imp:n=1  $ Lower Drum cavity
5   1 -1        20  -21  -24                u=2 imp:n=1  $ Upper Source
6   4 -7.85260  30  -32  -34 (-31:33:35)    u=2 imp:n=1  $ Upper CCC
7   0           31  -33  -35 (-20:21:24)    u=2 imp:n=1  $ Upper CCC cavity
8   0           #5 #6 #7                    u=2 imp:n=1  $ Upper Drum cavity
11  0           50  -51  -52      trcl=1 fill=1 imp:n=1  $ CCO #1
12  0           50  -51  -52      trcl=2 fill=1 imp:n=1  $ CCO #2
13  0           50  -51  -52      trcl=3 fill=1 imp:n=1  $ CCO #3
14  0           50  -51  -52      trcl=4 fill=1 imp:n=1  $ CCO #4
15  0           50  -51  -52      trcl=5 fill=1 imp:n=1  $ CCO #5
16  0           50  -51  -52      trcl=6 fill=1 imp:n=1  $ CCO #6
17  0           50  -51  -52      trcl=7 fill=1 imp:n=1  $ CCO #7
18  0           50  -51  -52      trcl=8 fill=2 imp:n=1  $ CCO #8
19  0           50  -51  -52      trcl=9 fill=2 imp:n=1  $ CCO #9
20  0           50  -51  -52     trcl=10 fill=2 imp:n=1  $ CCO #10
21  0           50  -51  -52     trcl=11 fill=2 imp:n=1  $ CCO #11
22  0           50  -51  -52     trcl=12 fill=2 imp:n=1  $ CCO #12
23  0           50  -51  -52     trcl=13 fill=2 imp:n=1  $ CCO #13
24  0           50  -51  -52     trcl=14 fill=2 imp:n=1  $ CCO #14
200 0           210 -250 -230
                #11 #12 #13 #14 #15 #16 #17
                #18 #19 #20 #21 #22 #23 #24    imp:n=1  $ ICV cavity
210 2 -8.01280  200 -210 -220                  imp:n=1  $ ICV/OCV bottom
220 2 -8.01280  250 -240 -220                  imp:n=1  $ ICV/OCV top
230 2 -8.01280  210 -250  230 -220             imp:n=4  $ ICV/OCV shell
240 3 -0.13215  110 -150 -130 (-200:220:240)   imp:n=8  $ OCA annulus
250 2 -8.01280  100 -110 -120                  imp:n=1  $ OCA bottom
260 2 -8.01280  150 -140 -120                  imp:n=1  $ OCA top
270 2 -8.01280  110 -150  130 -120             imp:n=16 $ OCA shell
777 0           120 -160  100 -140             imp:n=32 $ Outside to tally
888 0          -999 (-100:160:140)             imp:n=8  $ Outside pkg
999 0           999                            imp:n=0  $ Outside world

C
C Geometry Cards
C
```

```
10  pz 147.955                                          $ Lower Source bottom
11  pz 150.495                                          $ Lower Source top
14  c/z 0 0 1.27                                        $ Lower Source radius
20  pz 147.955                                          $ Upper Source bottom
21  pz 150.495                                          $ Upper Source top
24  c/z 0 0 1.27                                        $ Upper Source radius
C
30  pz 112.467                                          $ CCC base (outside)
31  pz 115.007                                          $ CCC base (inside)
32  pz 185.983                                          $ CCC lid (outside)
33  pz 183.443                                          $ CCC lid (inside)
34  cz 8.414                                            $ CCC shell (outside)
35  cz 7.703                                            $ CCC shell (inside)
C
50  pz 104.775                                          $ Drum bottom
51  pz 193.675                                          $ Drum top
52  cz 30.48                                            $ Drum radius (outside)
C
100 pz 0                                                $ OCA bottom (outside)
110 pz 0.635                                            $ OCA bottom (inside)
120 cz 119.38                                           $ OCA shell (outside)
130 cz 118.745                                          $ OCA shell (inside)
140 pz 306.07                                           $ OCA top (outside)
150 pz 305.435                                          $ OCA top (inside)
160 cz 319.38                                           $ 2-meter from OCA
C
200 pz 22.86                                            $ ICV/OCV bottom (outside)
210 pz 24.13                                            $ ICV/OCV bottom (inside)
220 cz 93.345                                           $ ICV/OCV shell (outside)
230 cz 92.234                                           $ ICV/OCV shell (inside)
240 pz 275.59                                           $ ICV/OCV top (outside)
250 pz 274.32                                           $ ICV/OCV top (inside)
C
777 pz 146.685                                          $ Ring tally (lower)
888 pz 151.765                                          $ Ring tally (upper)
999 sz 149.225 446.38                                   $ Outside world

C
C Translation Cards
C
*tr1 0 0 -44.609
*tr2 52.793 30.48 -44.609
*tr3 0 60.96 -44.609
*tr4 -52.793 30.48 -44.609
*tr5 -52.793 -30.48 -44.609
*tr6 0 -60.96 -44.609
*tr7 52.793 -30.48 -44.609
*tr8 0 0 44.609
*tr9 52.793 30.48 44.609
*tr10 0 60.96 44.609
*tr11 -52.793 30.48 44.609
*tr12 -52.793 -30.48 44.609
*tr13 0 -60.96 44.609
*tr14 52.793 -30.48 44.609
C
C Physics Cards
C
mode n
C
C Source Cards
C
C Cylindrical Sources
sdef pos=d1 erg=d2 par=1 axs=0 0 1 rad=d3 ext=d4
sc1 Cf252
C Source Position and Fraction
si1 L 0 0 104.616
      52.793 30.48 104.616
      0 60.96 104.616
      -52.793 30.48 104.616
      -52.793 -30.48 104.616
      0 -60.96 104.616
      52.793 -30.48 104.616
      0 0 193.834
      52.793 30.48 193.834
      0 60.96 193.834
      -52.793 30.48 193.834
      -52.793 -30.48 193.834
      0 -60.96 193.834
      52.793 -30.48 193.834
sp1   1 1 1 1 1 1 1 1 1 1 1 1 1 1
C Source Energy (MeV) and Fraction
```

```
si2 0.1 0.5 1.0 2.0 3.0
     4.0 6.0 8.0 10.0 15.0
sp2 0.00E+00 1.80E+11 2.71E+11 5.36E+11 4.10E+11
     2.73E+11 2.66E+11 8.53E+10 2.44E+10 8.36E+09
C Source Radius
si3 0 1.27
sp3 -21 1
C Source Extent
si4 -1.27 1.27
sp4 -21 0
C
C Tally Cards
C
f2:n 120
fc2 Surface Dose (mrem/hr)                          $ Surface detector
fs2 888 777 t
sd2 1.0 3810.44 1.0 1.0                             $ Surface tally area
f22:n 160
fc22 2-meter Dose (mrem/hr)                         $ 2-m detector
fs22 888 777 t
sd22 1.0 10194.144 1.0 1.0                          $ Surface tally area
C
C ANSI/ANS-6.1.1-1977 Neutron Flux to Dose Factors (mrem/hr)
de0 2.5e-08 1.0e-07 1.0e-06 1.0e-05 1.0e-04         $ Energy (Mev)
     1.0e-03 1.0e-02 1.0e-01 5.0e-01 1.0
     2.5 5.0 7.0 10.0 14.0 20.0
df0 3.67e-03 3.67e-03 4.46e-03 4.54e-03 4.18e-03    $ Conversion (mrem/hr)
     3.76e-03 3.56e-03 2.17e-02 9.26e-02 1.32e-01
     1.25e-01 1.56e-01 1.47e-01 1.47e-01 2.08e-01 2.27e-01
C
C Material Cards
C
m1   40000.60c -1.00                                $ Zirconium
m2   14000.60c -0.01                                $ Stainless Steel
     24052.60c -0.19
     25055.60c -0.02
     26057.60c -0.68
     28058.60c -0.10
m3   6000.60c  -0.60                                $ Urethane Foam
     7014.60c  -0.08
     8016.60c  -0.24
     1001.60c  -0.07
     14000.60c -0.01
mt3  poly.01t
m4   26000 -1.00                                    $ Carbon Steel (iron)
C
C Runtime and Print Cards
C
prdmp j j 1 2
ctme 30
```

6.1.2 HalfPACT Package Models

6.1.2.1 Carbon Steel CCCs with ^{60}Co Sources – dng001c.i

```
title HalfPACT CCO NCT Gamma Concentrated 1 g/cc Co60; CS CCC
C
C Cell Cards
C
1   1 -1        10  -11  -14                u=1 imp:p=1  $ Source
2   4 -7.85260  30  -32  -34 (-31:33:35)   u=1 imp:p=1  $ CCC
3   0           31  -33  -35 (-10:11:14)   u=1 imp:p=1  $ CCC cavity
4   0           #1 #2 #3                   u=1 imp:p=1  $ Drum cavity
11  0           50  -51  -52     trcl=1 fill=1 imp:p=1  $ CCO #1
12  0           50  -51  -52     trcl=2 fill=1 imp:p=1  $ CCO #2
13  0           50  -51  -52     trcl=3 fill=1 imp:p=1  $ CCO #3
14  0           50  -51  -52     trcl=4 fill=1 imp:p=1  $ CCO #4
15  0           50  -51  -52     trcl=5 fill=1 imp:p=1  $ CCO #5
16  0           50  -51  -52     trcl=6 fill=1 imp:p=1  $ CCO #6
17  0           50  -51  -52     trcl=7 fill=1 imp:p=1  $ CCO #7
200 0           210 -250 -230
                #11 #12 #13 #14 #15 #16 #17    imp:p=1  $ ICV cavity
210 2 -8.01280  200 -210 -220                  imp:p=1  $ ICV/OCV bottom
220 2 -8.01280  250 -240 -220                  imp:p=1  $ ICV/OCV top
230 2 -8.01280  210 -250  230 -220             imp:p=4  $ ICV/OCV shell
240 3 -0.13215  110 -150 -130 (-200:220:240)   imp:p=8  $ OCA annulus
250 2 -8.01280  100 -110 -120                  imp:p=1  $ OCA bottom
260 2 -8.01280  150 -140 -120                  imp:p=1  $ OCA top
270 2 -8.01280  110 -150  130 -120             imp:p=16 $ OCA shell
777 0           120 -160  100 -140             imp:p=32 $ Outside to tally
888 0          -999 (-100:160:140)             imp:p=8  $ Outside pkg
999 0           999                            imp:p=0  $ Outside world

C
C Geometry Cards
C
10  pz 109.855                                          $ Source bottom
11  pz 112.395                                          $ Source top
14  c/z 0 0 1.27                                        $ Source radius
C
30  pz 74.367                                           $ CCC base (outside)
31  pz 76.907                                           $ CCC base (inside)
32  pz 147.883                                          $ CCC lid (outside)
33  pz 145.343                                          $ CCC lid (inside)
34  cz 8.414                                            $ CCC shell (outside)
35  cz 7.703                                            $ CCC shell (inside)
C
50  pz 66.675                                           $ Drum bottom
51  pz 155.575                                          $ Drum top
52  cz 30.48                                            $ Drum radius (outside)
C
100 pz 0                                                $ OCA bottom (outside)
110 pz 0.635                                            $ OCA bottom (inside)
120 cz 119.38                                           $ OCA shell (outside)
130 cz 118.745                                          $ OCA shell (inside)
140 pz 229.87                                           $ OCA top (outside)
150 pz 229.235                                          $ OCA top (inside)
160 cz 319.38                                           $ 2-meter from OCA
C
200 pz 22.86                                            $ ICV/OCV bottom (outside)
210 pz 24.13                                            $ ICV/OCV bottom (inside)
220 cz 93.345                                           $ ICV/OCV shell (outside)
230 cz 92.234                                           $ ICV/OCV shell (inside)
240 pz 199.39                                           $ ICV/OCV top (outside)
250 pz 198.12                                           $ ICV/OCV top (inside)
C
777 pz 108.585                                          $ Ring tally (lower)
888 pz 113.665                                          $ Ring tally (upper)
999 sz 111.125 446.38                                   $ Outside world

C
C Translation Cards
C
*tr1 0 0 0
*tr2 52.793 30.48 0
*tr3 0 60.96 0
*tr4 -52.793 30.48 0
*tr5 -52.793 -30.48 0
```

```
*tr6 0 -60.96 0
*tr7 52.793 -30.48 0
C
C Physics Cards
C
mode p
C
C Source Cards
C
C Cylindrical Sources
sdef pos=d1 erg=d2 par=2 axs=0 0 1 rad=d3 ext=d4
sc1 Co60
C Source Position and Fraction
si1 L 0 0 111.125
      52.793 30.48 111.125
      0 60.96 111.125
      -52.793 30.48 111.125
      -52.793 -30.48 111.125
      0 -60.96 111.125
      52.793 -30.48 111.125
sp1   1 1 1 1 1 1 1
C Source Energy (MeV) and Intensity (%)
si2 L 3.469300E-01 8.262800E-01 1.173237E+00 1.332501E+00
     2.158770E+00 2.505000E+00
sp2 7.600000E-03 7.600000E-03 9.997360E+01 9.998560E+01
     1.110000E-03 2.000000E-06
C Source Radius
si3 0 1.27
sp3 -21 1
C Source Extent
si4 -1.27 1.27
sp4 -21 0
C
C Tally Cards
C
f2:p 120
fc2 Surface Dose (mrem/hr)                          $ Surface detector
fs2 888 777 t
sd2 1.0 3810.44 1.0 1.0                             $ Surface tally area
f22:p 160
fc22 2-meter Dose (mrem/hr)                         $ 2-m detector
fs22 888 777 t
sd22 1.0 10194.144 1.0 1.0                          $ Surface tally area
C
C ANSI/ANS-6.1.1-1977 Gamma Flux to Dose Factors (mrem/hr)
de0 1.0e-02 3.0e-02 5.0e-02 7.0e-02 1.0e-01         $ Energy (Mev)
     1.5e-01 2.0e-01 2.5e-01 3.0e-01 3.5e-01
     4.0e-01 4.5e-01 5.0e-01 5.5e-01 6.0e-01
     6.5e-01 7.0e-01 8.0e-01 1.00 1.40
     1.80 2.20 2.60 2.80 3.25
     3.75 4.25 4.75 5.00 5.25
     5.75 6.25 6.75 7.50 9.00
     11.00 13.00 15.0000
df0 3.96e-03 5.82e-04 2.90e-04 2.58e-04 2.83e-04    $ Conversion (mrem/hr)
     3.79e-04 5.01e-04 6.31e-04 7.59e-04 8.78e-04
     9.85e-04 1.08e-03 1.17e-03 1.27e-03 1.36e-03
     1.44e-03 1.52e-03 1.68e-03 1.98e-03 2.51e-03
     2.99e-03 3.42e-03 3.82e-03 4.01e-03 4.41e-03
     4.83e-03 5.23e-03 5.60e-03 5.80e-03 6.01e-03
     6.37e-03 6.74e-03 7.11e-03 7.66e-03 8.77e-03
     1.03e-02 1.18e-02 1.33e-02
C
C Material Cards
C
m1   40000 -1.00                                    $ Zirconium
m2   14000 -0.01                                    $ Stainless Steel
     24000 -0.19
     25000 -0.02
     26000 -0.68
     28000 -0.10
m3   6000  -0.60                                    $ Urethane Foam
     7000  -0.08
     8000  -0.24
     1000  -0.07
     14000 -0.01
m4   26000 -1.00                                    $ Carbon Steel (iron)
C
C Runtime and Print Cards
C
prdmp j j 1 2
ctme 30
```

6.1.2.2 Carbon Steel CCCs with ^{252}Cf Sources – dnn001c.i

```
title HalfPACT CCO NCT Neutron Concentrated 1 g/cc Cf252; CS CCC
C
C Cell Cards
C
1   1 -1        10  -11  -14                u=1 imp:n=1  $ Source
2   4 -7.85260  30  -32  -34 (-31:33:35)   u=1 imp:n=1  $ CCC
3   0           31  -33  -35 (-10:11:14)   u=1 imp:n=1  $ CCC cavity
4   0           #1 #2 #3                   u=1 imp:n=1  $ Drum cavity
11  0           50  -51  -52     trcl=1 fill=1 imp:n=1  $ CCO #1
12  0           50  -51  -52     trcl=2 fill=1 imp:n=1  $ CCO #2
13  0           50  -51  -52     trcl=3 fill=1 imp:n=1  $ CCO #3
14  0           50  -51  -52     trcl=4 fill=1 imp:n=1  $ CCO #4
15  0           50  -51  -52     trcl=5 fill=1 imp:n=1  $ CCO #5
16  0           50  -51  -52     trcl=6 fill=1 imp:n=1  $ CCO #6
17  0           50  -51  -52     trcl=7 fill=1 imp:n=1  $ CCO #7
200 0           210 -250 -230
                #11 #12 #13 #14 #15 #16 #17    imp:n=1  $ ICV cavity
210 2 -8.01280  200 -210 -220                  imp:n=1  $ ICV/OCV bottom
220 2 -8.01280  250 -240 -220                  imp:n=1  $ ICV/OCV top
230 2 -8.01280  210 -250  230 -220             imp:n=4  $ ICV/OCV shell
240 3 -0.13215  110 -150 -130 (-200:220:240)   imp:n=8  $ OCA annulus
250 2 -8.01280  100 -110 -120                  imp:n=1  $ OCA bottom
260 2 -8.01280  150 -140 -120                  imp:n=1  $ OCA top
270 2 -8.01280  110 -150  130 -120             imp:n=16 $ OCA shell
777 0           120 -160  100 -140             imp:n=32 $ Outside to tally
888 0          -999 (-100:160:140)             imp:n=8  $ Outside pkg
999 0           999                            imp:n=0  $ Outside world

C
C Geometry Cards
C
10  pz 109.855                                          $ Source bottom
11  pz 112.395                                          $ Source top
14  c/z 0 0 1.27                                        $ Source radius
C
30  pz 74.367                                           $ CCC base (outside)
31  pz 76.907                                           $ CCC base (inside)
32  pz 147.883                                          $ CCC lid (outside)
33  pz 145.343                                          $ CCC lid (inside)
34  cz 8.414                                            $ CCC shell (outside)
35  cz 7.703                                            $ CCC shell (inside)
C
50  pz 66.675                                           $ Drum bottom
51  pz 155.575                                          $ Drum top
52  cz 30.48                                            $ Drum radius (outside)
C
100 pz 0                                                $ OCA bottom (outside)
110 pz 0.635                                            $ OCA bottom (inside)
120 cz 119.38                                           $ OCA shell (outside)
130 cz 118.745                                          $ OCA shell (inside)
140 pz 229.87                                           $ OCA top (outside)
150 pz 229.235                                          $ OCA top (inside)
160 cz 319.38                                           $ 2-meter from OCA
C
200 pz 22.86                                            $ ICV/OCV bottom (outside)
210 pz 24.13                                            $ ICV/OCV bottom (inside)
220 cz 93.345                                           $ ICV/OCV shell (outside)
230 cz 92.234                                           $ ICV/OCV shell (inside)
240 pz 199.39                                           $ ICV/OCV top (outside)
250 pz 198.12                                           $ ICV/OCV top (inside)
C
777 pz 108.585                                          $ Ring tally (lower)
888 pz 113.665                                          $ Ring tally (upper)
999 sz 111.125 446.38                                   $ Outside world

C
C Translation Cards
C
*tr1 0 0 0
*tr2 52.793 30.48 0
*tr3 0 60.96 0
*tr4 -52.793 30.48 0
*tr5 -52.793 -30.48 0
*tr6 0 -60.96 0
*tr7 52.793 -30.48 0
C
C Physics Cards
C
```

```
mode n
C
C Source Cards
C
C Cylindrical Sources
sdef pos=d1 erg=d2 par=1 axs=0 0 1 rad=d3 ext=d4
sc1 Cf252
C Source Position and Fraction
si1 L 0 0 111.125
      52.793 30.48 111.125
      0 60.96 111.125
      -52.793 30.48 111.125
      -52.793 -30.48 111.125
      0 -60.96 111.125
      52.793 -30.48 111.125
sp1   1 1 1 1 1 1 1
C Source Energy (MeV) and Fraction
si2 0.1 0.5 1.0 2.0 3.0
     4.0 6.0 8.0 10.0 15.0
sp2 0.00E+00 1.80E+11 2.71E+11 5.36E+11 4.10E+11
     2.73E+11 2.66E+11 8.53E+10 2.44E+10 8.36E+09
C Source Radius
si3 0 1.27
sp3 -21 1
C Source Extent
si4 -1.27 1.27
sp4 -21 0
C
C Tally Cards
C
f2:n 120
fc2 Surface Dose (mrem/hr)                              $ Surface detector
fs2 888 777 t
sd2 1.0 3810.44 1.0 1.0                                 $ Surface tally area
f22:n 160
fc22 2-meter Dose (mrem/hr)                             $ 2-m detector
fs22 888 777 t
sd22 1.0 10194.144 1.0 1.0                              $ Surface tally area
C
C ANSI/ANS-6.1.1-1977 Neutron Flux to Dose Factors (mrem/hr)
de0 2.5e-08 1.0e-07 1.0e-06 1.0e-05 1.0e-04             $ Energy (Mev)
     1.0e-03 1.0e-02 1.0e-01 5.0e-01 1.0
     2.5 5.0 7.0 10.0 14.0 20.0
df0 3.67e-03 3.67e-03 4.46e-03 4.54e-03 4.18e-03        $ Conversion (mrem/hr)
     3.76e-03 3.56e-03 2.17e-02 9.26e-02 1.32e-01
     1.25e-01 1.56e-01 1.47e-01 1.47e-01 2.08e-01 2.27e-01
C
C Material Cards
C
m1   40000.60c -1.00                                    $ Zirconium
m2   14000.60c -0.01                                    $ Stainless Steel
     24052.60c -0.19
     25055.60c -0.02
     26057.60c -0.68
     28058.60c -0.10
m3   6000.60c  -0.60                                    $ Urethane Foam
     7014.60c  -0.08
     8016.60c  -0.24
     1001.60c  -0.07
     14000.60c -0.01
mt3  poly.01t
m4   26000 -1.00                                        $ Carbon Steel (iron)
C
C Runtime and Print Cards
C
prdmp j j 1 2
ctme 30
```